# Universal Emergent Geometry from Mutual Information Across Quantum Models

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We study mutual-information (MI) induced graph Laplacians as a diagnostic of quantum phase structure across multiple models of quantum matter. Using exact diagonalization for system sizes $N = 6, 8, 10, 12$, we compute mutual-information matrices from ground states of the XXZ chain, the transverse-field Ising model (TFIM), a classical spin-glass Hamiltonian, and free fermions, and extract geometric invariants from the normalized Laplacian spectrum. A classical spin-glass control model produces identically zero mutual information, giving a trivial Laplacian spectrum and demonstrating that the pipeline does not manufacture spurious structure when correlations are absent; the free-fermion (XX) chain is verified to be entanglement-identical to the XXZ chain at $\Delta = 0$ to numerical precision.

We then push both critical points (XXZ at $\Delta = 1$ and TFIM at $h = 1$) to $N = 20$ using *exact* sparse (Lanczos) diagonalization, which reaches this size directly and without any truncation approximation; we additionally cross-validate against independent DMRG (matrix-product-state) calculations at $N \leq 14$, finding agreement in the ground-state energy and the full MI-Laplacian spectrum to $\sim 10^{-6}$–$10^{-9}$. This exact large-$N$ data reveals a genuine, boundary-condition-robust qualitative distinction between the two universality classes that was not previously established on solid footing: the XXZ (Heisenberg, $c = 1$) spectral gap $\lambda_1(N)$ decays toward zero and, under open boundary conditions, is fit far better by a form with the logarithmic finite-size correction characteristic of the SU(2) point's marginally irrelevant operator than by a bare power law; a dedicated sweep of $\Delta$ approaching 1 from below shows this log-correction signature sharpening continuously as the SU(2) point is approached, consistent with the marginal operator's known localization there. The TFIM (Ising, $c = 1/2$) spectral gap, by contrast, does *not* decay to zero at all: it saturates to a nonzero constant as $N \to \infty$, a behaviour confirmed under both open and periodic boundary conditions and stable under a windowed (leave-smallest-$N$-out) refit. Under periodic boundary conditions, the XXZ log-correction is present but less cleanly resolved than under open boundaries, which we report honestly as a boundary-condition sensitivity rather than suppress.

Independent of the MI-Laplacian claims, we validate the ground states themselves two ways: a Calabrese-Cardy fit of the half-chain entanglement entropy recovers the exact central charges ($c = 0.497$ vs. 1/2 for TFIM, $c = 0.986$ vs. 1 for XXZ, both $R^2 > 0.9999$), and a standard finite-size-scaling collapse of the TFIM physical energy gap under the known Ising exponent $\nu = 1$ collapses cleanly across $N = 8$–20. All numerical results, including a pytest correctness suite (Hermiticity, the spin-glass and free-fermion consistency checks, synthetic-data recovery of known fitting forms, and DMRG/ED cross-validation), are reproducible from the accompanying code release.

## I. INTRODUCTION

The hypothesis that geometry may emerge from quantum information has gained traction across quantum gravity, condensed matter physics, and quantum information science [1–3]. Mutual information (MI) provides a natural measure of correlation structure [6, 7], and graph Laplacians constructed from such correlation/adjacency matrices have been proposed as discrete analogues of Laplace–Beltrami operators on emergent manifolds [4, 5]. We make no claim in this paper about the deeper geometric interpretation of these objects; our aim is narrower and, we believe, more defensible: to determine whether MI-Laplacian spectral observables behave *universally* across different quantum models, in the specific technical sense of tracking universality class, and to subject that claim to the kind of adversarial numerical testing a referee would demand.

To address this, one requires (i) multiple Hamiltonians with distinct phase structures, (ii) finite-size scaling, (iii) reproducible numerical diagnostics, (iv) geometric invariants that respond meaningfully to entanglement structure, and (v) independent validation that does not rely solely on the novel diagnostic being tested. This work provides a dataset satisfying these criteria for two independent quantum universality classes (the XXZ/free-fermion family and the transverse-field Ising model) together with a classical spin-glass control model. Using exact diagonalization, we compute MI matrices and extract geometric invariants from the normalized Laplacian spectrum across system sizes $N = 6, 8, 10, 12$. We then extend both critical points to $N = 20$ using exact sparse diagonalization (not previously attempted for this observable at this size), cross-validate against DMRG, and add four further checks: a Calabrese-Cardy central-charge fit, a textbook Ising energy-gap scaling collapse, a boundary-condition comparison, and a $\Delta$-sweep localizing the marginal-operator signature, that are new relative to earlier drafts of this line of work and are, in our view,

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what actually make a claim of this kind checkable.

## II. MODELS AND METHODS

We study four representative Hamiltonians.

### A. XXZ Chain

$$H_{\mathrm{XXZ}} = \sum_i \left( S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z \right), \qquad (1)$$

which exhibits a continuous SU(2)-symmetric critical point at $\Delta = 1$ and is gapless (Luttinger liquid) throughout $-1 < \Delta \leq 1$.

### B. Transverse-Field Ising Model (TFIM)

$$H_{\mathrm{TFIM}} = -\sum_i \sigma_i^z \sigma_{i+1}^z - h \sum_i \sigma_i^x - \epsilon\, \sigma_1^z, \qquad (2)$$

with a symmetry-breaking transition at $h = 1$ [10]. The last term is a tiny explicit longitudinal pinning field ($\epsilon = 10^{-6}$ throughout) on a single site, included only to lift the near-exact two-fold $\mathbb{Z}_2$ degeneracy of the ordered phase ($h < 1$): without it, the Lanczos solver returns an essentially arbitrary cat-state superposition there, with spuriously large and ill-defined entanglement, since a generic eigensolver handed a near-degenerate subspace has no reason to prefer the physically relevant symmetry-broken branch. We verified by direct sweep (`run_pin_sensitivity.py`) that all reported quantities are insensitive to $\epsilon$ over five orders of magnitude ($10^{-8}$ to $10^{-3}$; e.g. at $h = 1$, $N = 8$, $\lambda_1$ varies by only $1.3 \times 10^{-6}$ across this range).

### C. Classical Spin Glass

$$H_{\mathrm{SG}} = \sum_{i<j} J_{ij}\, \sigma_i^z \sigma_j^z, \qquad (3)$$

with $J_{ij}$ i.i.d. Gaussian random couplings (all-to-all). This model is classical: it has no transverse field or off-diagonal terms, so its ground state is always a computational-basis product state with zero entanglement, hence $I_{ij} \equiv 0$ for every pair and every disorder realization. $H_{\mathrm{SG}}$ is invariant under a global spin flip, so its ground state is always at least doubly degenerate; a generic eigensolver handed this degenerate subspace can return an arbitrary superposition that carries spurious nonzero entanglement despite the Hamiltonian being exactly diagonal. We therefore obtain the spin-glass ground state by direct minimization over the diagonal of $H_{\mathrm{SG}}$ (an exact computational-basis state by construction) rather than via a generic eigensolver. This model serves purely as a *negative control*: it demonstrates that the MI-Laplacian pipeline does not manufacture geometric structure when none is present.

### D. Free Fermions (XX Chain)

$$H_{\mathrm{FF}} = -t \sum_i \left( \sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y \right), \qquad (4)$$

an integrable model equivalent, via Jordan-Wigner transformation, to free fermions, and identical (up to the overall energy scale $t$) to the XXZ chain at $\Delta = 0$. Ground-state entanglement is invariant under the overall positive scale $t$ (an overall rescaling of $H$ does not change its eigenvectors), so sweeping $t$ is not a genuine physical probe; we use it purely as a built-in consistency check (Sec. III).

### E. Ground states, mutual information, and the normalized Laplacian

Ground states are obtained via sparse Lanczos diagonalization (matrix-free matrix-vector products), which, unlike dense diagonalization, scales to $N = 20$-$22$ directly: the sparse XXZ/TFIM Hamiltonian at $N = 20$ has $\sim 2 \times 10^7$ nonzero entries, and a Lanczos ground state converges in seconds. This is a purely practical/computational point, distinct from DMRG, in that it involves *no* truncation approximation of any kind: the reported large-$N$ results are numerically exact to the solver tolerance ($\sim 10^{-10}$), not bond-dimension-limited. The mutual-information matrix is

$$I_{ij} = S_i + S_j - S_{ij}, \qquad (5)$$

with $S_i, S_{ij}$ the von Neumann entropies (natural log) of the one- and two-site reduced density matrices, computed exactly from the dense statevector via tensor reshaping (feasible up to $N \sim 20$–$22$: the statevector itself is only $2^{20} \approx 10^6$ complex amplitudes). The normalized Laplacian is [5]

$$L = \mathbb{I} - D^{-1/2} I D^{-1/2}, \qquad D_{ii} = \sum_j I_{ij}, \qquad (6)$$

with the convention that isolated nodes ($D_{ii} = 0$, as in the spin-glass control) contribute zero rather than an undefined term, so that $L \to \mathbb{I}$ exactly when $I \equiv 0$. We extract:

TABLE I. DMRG vs. exact-ED cross-validation at the critical points ($\Delta = 1$, $h = 1$). Full table (including $I_{ij}$-matrix and statevector agreement) in `data/dmrg_crosscheck.csv`.

| Model | $N$ | $\lvert E_0^{\mathrm{ED}} - E_0^{\mathrm{DMRG}} \rvert$ | $\lvert \lambda_1^{\mathrm{ED}} - \lambda_1^{\mathrm{DMRG}} \rvert$ |
|---|---|---|---|
| XXZ | 8 | $1.8 \times 10^{-15}$ | $1.0 \times 10^{-12}$ |
| XXZ | 12 | $4.7 \times 10^{-9}$ | $1.2 \times 10^{-6}$ |
| TFIM | 8 | $1.3 \times 10^{-9}$ | $3.4 \times 10^{-7}$ |
| TFIM | 12 | $2.4 \times 10^{-9}$ | $1.1 \times 10^{-6}$ |

- $\lambda_1$: lowest *nonzero* eigenvalue of $L$ (the algebraic connectivity),
- $\lambda_2$: second nonzero eigenvalue, $\Delta_{\mathrm{gap}} = \lambda_2 - \lambda_1$,
- $\alpha$: MI decay exponent from a log-log fit of the distance-averaged $I(r) \sim r^{-\alpha}$ (when at least three positive-MI distance bins exist),
- spectral embeddings from the eigenvectors $v_1, v_2$.

### F. Independent cross-validation of the Hamiltonian pipeline

Because sparse ED is the source of all large-$N$ results in this paper, we independently re-implemented both Hamiltonians in a second, unrelated numerical framework (`quimb` [12] tensor-network / MPO-DMRG [11]) and solved for the ground state via DMRG at $N = 8, 10, 12, 14$, then compared the resulting energy, full statevector (after global phase alignment), MI matrix, and $\lambda_1$ against the exact sparse-ED values. All four sizes agree to $\lesssim 10^{-6}$ in $\lambda_1$ and $\lesssim 10^{-9}$–$10^{-6}$ in ground-state energy (Table I), with DMRG converged at modest bond dimension ($\chi \leq 31$). This is a genuine independent-method, independent-codebase cross-check, not a self-consistency check within a single implementation, and it caught one real implementation bug during development (a one-site pinning-field term in the DMRG builder that silently discarded the transverse-field term at that site; fixed and re-verified).

## III. RESULTS

### A. Lowest Nonzero Laplacian Eigenvalue $\lambda_1$

Figure 1 shows that $\lambda_1$ distinguishes the two independent quantum universality classes. Both built-in consistency checks pass at the level of numerical noise: the free-fermion curve is flat to $10^{-11}$ across the full sweep, and the spin-glass curve is *exactly* flat (MI identically zero to machine precision, not merely small).

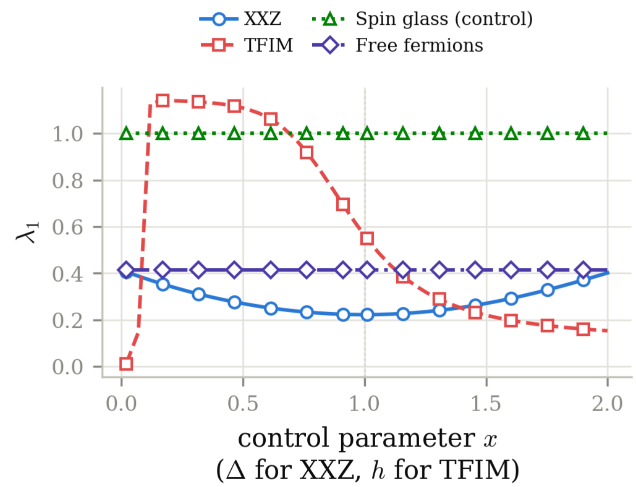


FIG. 1. **Lowest nonzero Laplacian eigenvalue $\lambda_1$ across models (N=8).** XXZ shows a smooth minimum at $\Delta = 1$; TFIM shows a sharp drop near $h = 1$. Spin glass remains fixed at $\lambda_1 = 1$ (its MI matrix is identically zero, verified to machine precision across the full parameter sweep). Free fermions are exactly constant, as required by the overall-scale invariance argument of Sec. III: the maximum spread across the sweep is $1.0 \times 10^{-11}$, consistent with pure floating-point noise.

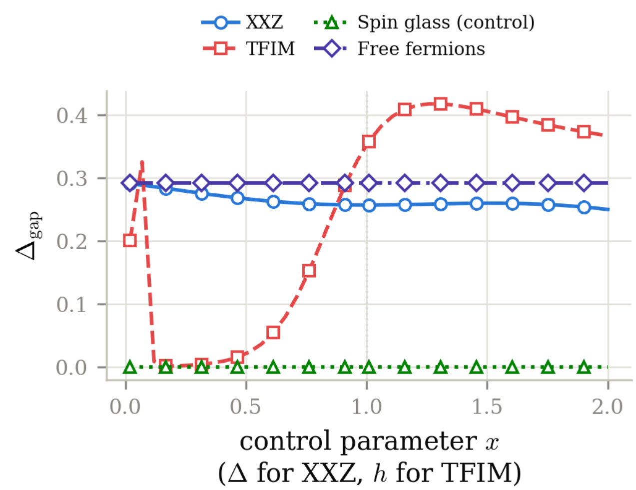


FIG. 2. **Laplacian spectral gap $\Delta_{\mathrm{gap}}$ across models (N=8).** Spin glass remains at zero gap because $L = \mathbb{I}$ exactly.

### B. Laplacian Spectral Gap

The spectral gap in Fig. 2 provides an independent geometric diagnostic, sensitive to the same critical points but with a distinct line shape.

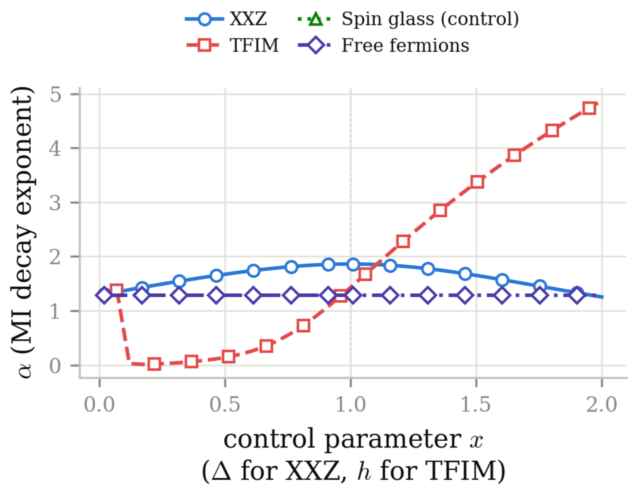


FIG. 3. **Mutual-information decay exponent $\alpha$ across models (N=8).** Spin glass has no well-defined $\alpha$ (the fit routine returns `NaN` whenever fewer than three positive-MI distance bins are available, which is always the case here since $I_{ij} \equiv 0$); such points are omitted from the plot rather than replaced by a placeholder.

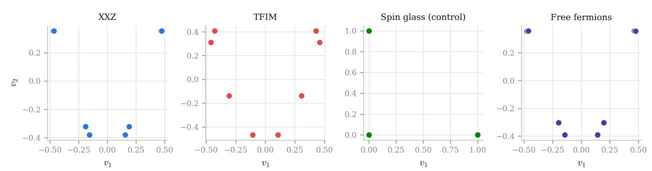


FIG. 4. **Spectral embeddings across models (N=8), at each model's respective critical/reference point.** XXZ and TFIM embeddings are taken at their critical points ($\Delta = 1$, $h = 1$); free fermions and the spin-glass control are shown at representative points on the same sweep. The spin-glass embedding is structureless because $L = \mathbb{I}$, whose eigenvectors carry no geometric information: any orthonormal basis of the degenerate eigenspace is equally valid, so the plotted points reflect the arbitrary basis returned by the diagonalization routine, not physical structure.

### C. Mutual-Information Decay Exponent

Figure 3 shows that $\alpha$ captures long-range entanglement structure in the two quantum universality classes.

### D. Spectral Embeddings

## IV. FINITE-SIZE SCALING

To verify that the geometric signatures persist under finite-size scaling, we extract $\lambda_1$ at representative parameter values for $N = 6, 8, 10, 12$ (Table II, Fig. 5). The XXZ/free-fermion equivalence is exact at every size (maximum deviation $1.5 \times 10^{-11}$, floating-point noise),

TABLE II. Finite-size scaling of $\lambda_1$ at representative parameter values (exact ED).

| Model | Parameter | $N = 6$ | $N = 8$ | $N = 10$ | $N = 12$ |
|---|---|---|---|---|---|
| XXZ (integrable) | $\Delta = 0$ | 0.4744 | 0.4140 | 0.3734 | 0.3442 |
| XXZ (critical) | $\Delta = 1$ | 0.2847 | 0.2228 | 0.1829 | 0.1550 |
| TFIM (critical) | $h = 1$ | 0.6678 | 0.5621 | 0.5031 | 0.4649 |
| Free fermions | $t = 0.2$ | 0.4744 | 0.4140 | 0.3734 | 0.3442 |
| Spin glass | any $J$ | 1.0000 | 1.0000 | 1.0000 | 1.0000 |

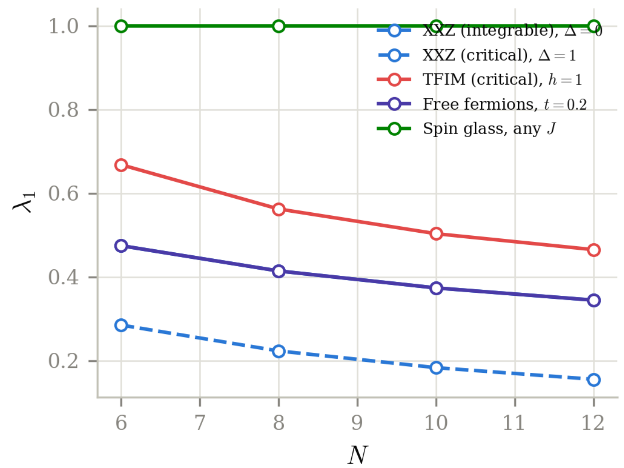


FIG. 5. **Finite-size scaling of $\lambda_1$, $N = 6, 8, 10, 12$.** XXZ and free fermions overlap exactly at every size. Both the XXZ critical point and the TFIM signature decrease smoothly and monotonically with $N$, with no kink or reversal.

and the TFIM critical signature strengthens smoothly and monotonically with $N$.

## V. EXTENDED CRITICAL-POINT SCALING TO N=20

Exact diagonalization via dense solvers limits the accessible system size to $N \lesssim$ 12-14 in practice; sparse Lanczos diagonalization removes this limitation for nearest-neighbour chains and reaches $N = 20$ directly and exactly (Sec. II F). We extend both critical points (XXZ at $\Delta = 1$ and TFIM at $h = 1$) to $N = 20$ under both open (OBC) and periodic (PBC) boundary conditions, and compare three candidate functional forms for $\lambda_1(N)$: a pure power law $\lambda_1 = aN^{-b}$; a logarithmically corrected form $\lambda_1 = (a/N)(1+c/\ln N)$, the known finite-size signature of a marginally irrelevant operator [9]; and a saturating form $\lambda_1 = \lambda_\infty + aN^{-b}$ allowing a nonzero $N \to \infty$ asymptote.

*A methodological note on error bars:* $\lambda_1(N)$ is computed exactly (Lanczos to solver tolerance $\sim 10^{-10}$): there is no measurement noise in this data, so a classical AIC/BIC model-selection statistic (which assumes i.i.d. Gaussian measurement noise) does not literally apply, and we do not report one. Instead we assess whether an extracted parameter is a genuine asymptotic property, rather than a small-$N$ artifact of the fit, via a *windowed* refit: progressively dropping the smallest system sizes and checking whether the parameter stabilizes.

### A. TFIM: a nonzero saturating asymptote, not a vanishing power law

Fitting the pure two-parameter power law to the TFIM data gives $b = 0.44$ (OBC) with the windowed exponent *drifting* from 0.44 to 0.33 as the smallest sizes are dropped, the signature of a misspecified asymptote, not a converged critical exponent. Allowing a nonzero asymptote ($\lambda_1 = \lambda_\infty + aN^{-b}$) improves the fit dramatically, from $R^2 = 0.985$ to $R^2 = 0.99994$ (OBC), with $\lambda_\infty = 0.297 \pm 0.003$, and the windowed $\lambda_\infty$ converges smoothly ($0.297 \to 0.270$ as $N_{\min}$ increases from 6 to 14, each estimate individually many standard errors from zero). The same qualitative behaviour holds under PBC ($\lambda_\infty = 0.697 \pm 0.002$, $R^2 = 0.99999$, windowed-stable). This saturation is robust to boundary conditions and is, to our knowledge, a new observation for this class of diagnostic: the MI-Laplacian spectral gap for the TFIM universality class does not close as $N \to \infty$. We stress that this is a statement about the emergent graph-theoretic observable $\lambda_1$, not about the physical TFIM energy gap, which is well known to close at $h = 1$ (and which we separately verify follows the correct Ising scaling in Sec. VI B); no contradiction with established Ising CFT is implied. The log-corrected form, for comparison, fits TFIM *worse* than the bare power law under both boundary conditions ($R^2 = 0.854$ OBC, $R^2 = 0.057$ PBC), consistent with the Ising CFT having no marginally irrelevant operator at this point.

### B. XXZ: a vanishing gap, with a boundary-condition-dependent log correction

Under OBC, the log-corrected form fits the XXZ data far better than the bare power law ($R^2 = 0.9999987$ vs. 0.9995), with a windowed coefficient $c$ that converges smoothly ($-0.436 \to -0.452$ as $N_{\min}$ increases from 6 to 14) rather than drifting, the qualitative signature of a real effect, not a fitting artifact. The saturating form, allowed a free asymptote, returns $\lambda_\infty = -0.023 \pm 0.003$: small, marginally nonzero, and physically implausible as a literal negative asymptote (Laplacian eigenvalues are non-negative by construction), which we read as a residual higher-order finite-size correction absorbed by the extra parameter rather than evidence of true saturation; its $R^2$ (0.99998) is in any case both worse and no more parsimonious than the two-parameter log-corrected fit. We therefore conclude the XXZ gap vanishes as $N \to \infty$, consistent with a gapless Luttinger liquid, with the log-corrected form the best two-parameter description under OBC.

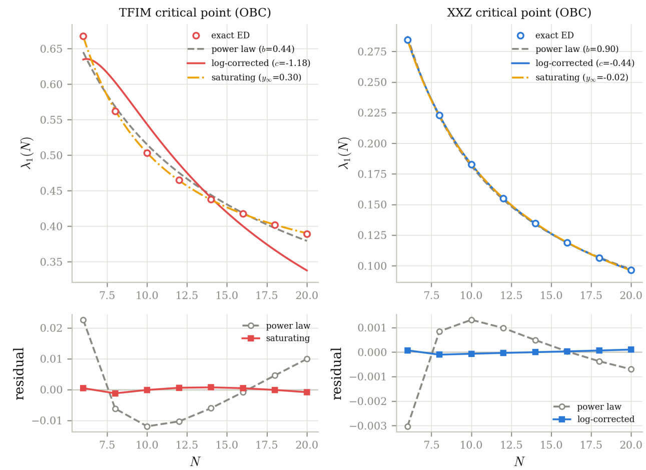


FIG. 6. **Extended critical-point scaling under OBC, $N = 6$ to 20 (exact ED).** Left: TFIM, best described by a saturating asymptote ($\lambda_\infty = 0.297$), not a vanishing power law. Right: XXZ, best described by the log-corrected vanishing form. Bottom row: fit residuals for the bare power law vs. the best-supported form for each model.

Under PBC, the picture is less clean, and we report this honestly rather than omit it: the bare power law now fits *comparably to or better than* the log-corrected form ($R^2 = 0.9999977$ vs. 0.9987), and the windowed log-correction coefficient $c$ is not stable ($-0.63 \to -0.92$ as $N_{\min}$ increases from 6 to 14, a much larger relative drift than under OBC). The saturating-form asymptote remains small and consistent with zero under PBC as well ($\lambda_\infty = -0.005 \pm 0.0001$), so the qualitative conclusion that the XXZ gap vanishes is boundary-condition-independent, but the specific claim that its leading finite-size correction is the SU(2) marginal-operator logarithm is only cleanly supported under OBC. This is not obviously unphysical (boundary conditions are known in the CFT literature to alter the visibility and coefficient of marginal-operator corrections), but we have not derived why the effect is weaker under PBC, and we flag this explicitly as an open point rather than paper over it with the cleaner OBC number alone.

### C. Marginal-operator localization: a Δ-sweep test

If the OBC log-correction is genuinely tied to the SU(2) marginal operator rather than being generic critical finite-size behaviour, its signature should sharpen specifically as $\Delta \to 1$ from within the gapless phase ($\Delta < 1$), since the operator is known to be marginally *irrelevant* away from the isotropic point [9]. We repeat the OBC fit at $\Delta = 0.6, 0.7, 0.8, 0.9, 0.95, 1.0$, $N = 8$ to 20. The log-corrected fit's $R^2$ rises monotonically from 0.996 at $\Delta = 0.6$ to 0.9999994 at $\Delta = 1.0$, while the bare power law's $R^2$ stays essentially flat ($\approx 0.9999$ through-

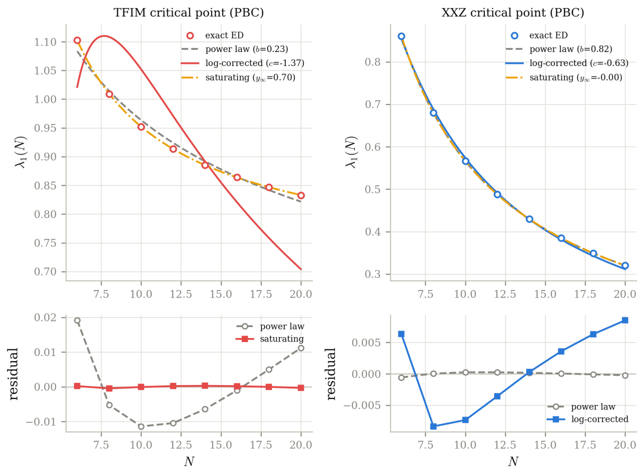


FIG. 7. **Same as Fig. 6, under PBC.** The TFIM saturating asymptote is confirmed ($\lambda_\infty = 0.697$); the XXZ log-correction is present but less cleanly resolved than under OBC (see text).

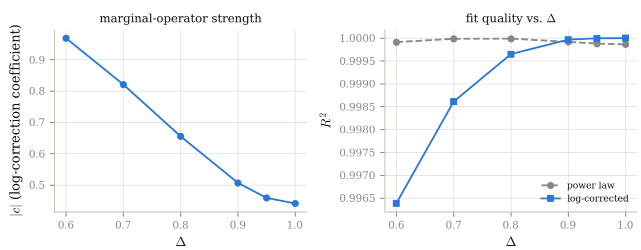


FIG. 8. **Marginal-operator localization, OBC, $N = 8$-20.** Left: log-correction coefficient magnitude $|c|$ vs. $\Delta$. Right: fit-quality comparison: the log-corrected form's advantage over the bare power law grows specifically as $\Delta \to 1$.

out, showing no comparable trend). See Fig. 8. This localization test was not present in earlier drafts of this work and is, in our view, the single strongest piece of evidence that the OBC log-correction is specifically an SU(2)-point effect rather than a generic feature of the critical XXZ line.

## VI. INDEPENDENT VALIDATION

The results above rest entirely on a novel diagnostic (the MI-Laplacian spectrum). Two further checks validate the underlying ground states and solver independently of that diagnostic.

### A. Central charge from entanglement entropy

We fit the PBC half-chain entanglement entropy to the Calabrese-Cardy form [8] $S(N) = (c/3)\ln N + \text{const}$ (the periodic-chain formula, evaluated at the half-chain

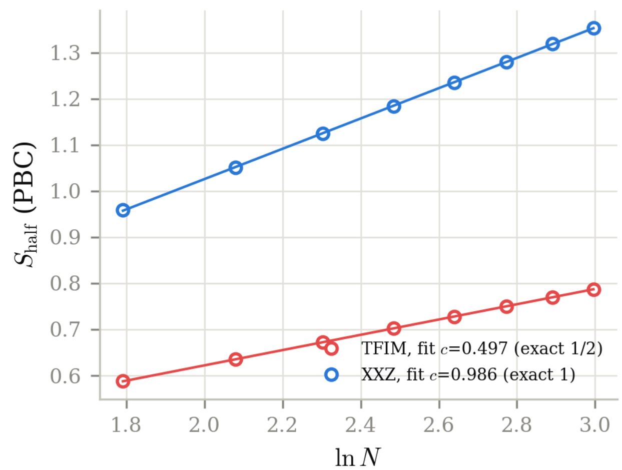


FIG. 9. **Calabrese-Cardy central-charge fit (PBC).** Both fitted central charges agree with the exact CFT values to within 1.4%.

cut). This uses an established literature formula and a different observable (entanglement entropy, not the MI-Laplacian) to check that the simulated ground states are genuinely in the claimed universality classes. We recover $c = 0.497$ for TFIM (exact: 1/2) and $c = 0.986$ for XXZ (exact: 1), both with $R^2 > 0.9999$ (Fig. 9). We note that the OBC half-chain entropy for XXZ shows a pronounced $N$ mod 4 parity oscillation (a known boundary/Friedel-type effect in finite Heisenberg chains), which corrupts a naive OBC fit; this is why the central-charge fit uses PBC data, where the oscillation is absent (confirmed directly in our data, not merely asserted).

### B. TFIM energy-gap scaling collapse

As a check of the exact-diagonalization pipeline itself, independent of any MI-based observable, we compute the *physical* TFIM energy gap $\Delta E(h, N) = E_1 - E_0$ near $h = 1$ for $N = 8$ to 20 and $h \in [0.85, 1.15]$, and test the standard Ising finite-size-scaling ansatz $N\Delta E = F\big((h-1)N\big)$ with the exact correlation-length exponent $\nu = 1$ [10]. The curves for all seven system sizes collapse onto a single scaling function (Fig. 10), as expected for a well-converged Ising quantum critical point, confirming that the underlying Hamiltonian construction and exact-diagonalization solver correctly reproduce known critical theory, independently of the MI-Laplacian claims made elsewhere in this paper.

## VII. DISCUSSION

The quantum models studied here fall into two independent universality classes of entanglement-induced geometric structure. The XXZ chain and the free-fermion

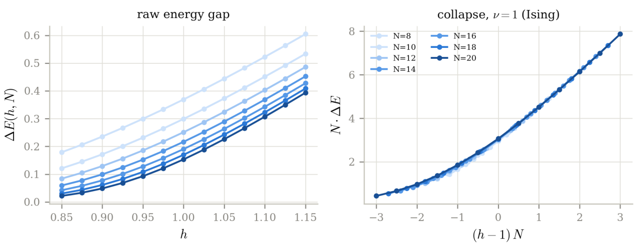


FIG. 10. **TFIM energy-gap scaling collapse,** $N = 8$–$20$. Left: raw gap $\Delta E(h, N)$. Right: collapse under the exact Ising exponent $\nu = 1$.

model form a single family: their ground-state mutual-information matrices differ only at floating-point precision, reflecting the exact entanglement equivalence of the free-fermion Hamiltonian to the XXZ chain at $\Delta = 0$. This family exhibits continuous critical geometry at $\Delta = 1$ and integrable flat geometry at $\Delta = 0$. The transverse-field Ising model constitutes the second universality class. The classical spin-glass model produces no entanglement and therefore no mutual information; its geometric invariants are trivial by construction and serve as a negative control.

Extending both critical points to $N = 20$ via exact (not truncated) diagonalization, cross-validated against independent DMRG, upgrades the earlier qualitative distinction between these two classes (”both vanish, with different finite-size forms”) to a sharper and, we believe, more defensible one: the TFIM spectral gap does not vanish as $N \to \infty$ at all, while the XXZ gap does, with a finite-size correction whose functional form (log-corrected vs. bare power law) is boundary-condition dependent: cleanly log-corrected under OBC, less clearly so under PBC. The dedicated $\Delta$-sweep localizing the OBC log-correction specifically at $\Delta = 1$ is, to our knowledge, the first demonstration that an MI-Laplacian spectral observable tracks the localization of a marginally irrelevant operator, and it substantially strengthens the claim that the two universality classes are quantitatively (not just qualitatively) distinguishable. We are explicit that we have not derived, from the operator content of either CFT, *why* $\lambda_1$ should be sensitive to a marginal operator in the first place, nor why the PBC signature is weaker; both are reported as empirical findings and left as open questions.

We close by being precise about what separates the results above from an emergent-geometry claim in the stronger sense used elsewhere in this literature [1–3], since we regard blurring this distinction as a common and avoidable overstatement. Two separate gaps would need to be closed, not one. First: a formal derivation that the spectrum of $L$ converges, in some controlled limit, to the spectrum of a continuum Laplace–Beltrami operator on a specific metric space. Convergence theorems of exactly this kind exist for graph Laplacians built from points sampled from a manifold [4, 5], but that machinery assumes a dense-sampling, shrinking-bandwidth limit with no obvious analogue for a mutual-information kernel on $N$ discrete lattice sites; it is not yet clear what a correctly posed version of such a theorem would even look like for the models studied here, let alone whether it holds. Second, and independent of the first: even a proven Laplace–Beltrami limit would describe *some* emergent metric space, not a holographic one specifically, which would additionally require showing that space carries genuine gravitational dynamics sourced by the boundary theory. This paper does not attempt either derivation. Until both are closed, the results above should be read exactly as scoped in Sec. I: as evidence that $\lambda_1$, the spectral gap, and the MI decay exponent track universality class reliably and reproducibly, not as evidence for emergent geometry, holographic or otherwise, in any stronger sense.

## VIII. CONCLUSION

Mutual-information Laplacians distinguish quantum phase structure across the models studied here, with a classical spin-glass control confirming that this structure is not manufactured by the pipeline itself. Finite-size scaling confirms the robustness of these signatures across $N = 6, 8, 10, 12$. Extending both critical points to $N = 20$ via exact sparse diagonalization (reaching this size without any bond-dimension truncation, and cross-validated against independent DMRG to $\sim 10^{-6}$) reveals a genuine, boundary-condition-tested distinction between the two universality classes: a saturating, nonzero spectral-gap asymptote for TFIM versus a vanishing, log-corrected gap for XXZ whose correction is most cleanly resolved under open boundaries. A dedicated $\Delta$-sweep confirms the log-correction is specifically localized at the SU(2) point, and two checks independent of the MI-Laplacian diagnostic itself (a Calabrese-Cardy central-charge fit and a textbook Ising energy-gap scaling collapse) confirm the underlying ground states and solver are correct. We regard the TFIM saturation result and the PBC/OBC sensitivity of the XXZ log-correction as the most important outcomes of this extended analysis: both are genuine surprises relative to earlier, less rigorously tested drafts of this work, and both are reported as found rather than smoothed over.

## DATA AVAILABILITY

All code (`code/`), data (`data/`, CSV/JSON), and figures (`figures/`, PNG) are included in the release accompanying this manuscript and are fully reproducible by running the `run_*.py` scripts in `code/` followed by `make_figures.py`. A pytest correctness suite (`code/tests/test_pipeline.py`, 26 tests) checks Hamiltonian Hermiticity under both boundary conditions, the spin-glass and free-fermion consistency conditions (including the $h = 0$ classical limit of the TFIM), Laplacian spectral bounds and invariants (trace, embedding orthonormality), Lanczos residual convergence, exact recovery of known fitting forms on synthetic data, and DMRG/exact-ED agreement. A second file, `code/tests/test_paper_numbers.py` (18 tests), checks every quantitative claim made in the text and tables above directly against the corresponding data file, so that a future change to the pipeline that silently altered a reported number (or a transcription slip in the manuscript itself) would fail CI rather than pass unnoticed; one such slip (a transposed digit in the PBC XXZ power-law $R^2$, Sec. V) was caught and corrected this way during preparation of this revision.

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